WORKED SOLUTION

STEP I 2013, Question 1

Pure • Equations with Square Roots

> **STEP I 2013, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/13-s1-q1

The key idea

Each equation is a quadratic in disguise. Call the square root y, write everything else in terms of y, and you get an ordinary quadratic in y. The one thing to watch is that a square root is never negative, so $y \geq 0$: that condition is what throws away the spurious root each time.

Solution

Part (i)

With $y = \sqrt{x}$, $y \geq 0$, we have $x = y^2$, and the equation becomes

$$y^2 + 3y - \tfrac{1}{2} = 0 \quad \Longleftrightarrow \quad 2y^2 + 6y - 1 = 0.$$

By the quadratic formula,

$$y = \frac{-6 \pm \sqrt{36+8}}{4} = \frac{-3 \pm \sqrt{11}}{2}.$$

The minus sign gives a negative value, which is impossible for a square root. Since $\sqrt{11} > 3$, the plus sign gives $y = \frac{1}{2}(\sqrt{11} - 3) > 0$, so this is the only option. Then

$$x = y^2 = \frac{11 - 6\sqrt{11} + 9}{4} = \frac{10 - 3\sqrt{11}}{2}.$$

Every step is reversible once we know $y \geq 0$, so this is indeed the (only) real root.

Part (ii)(a)

Here the square root is of $x+2$, so we put $y = \sqrt{x+2}$, $y \geq 0$, giving $x = y^2 - 2$. The equation becomes

$$y^2 - 2 + 10y - 22 = 0 \quad \Longleftrightarrow \quad y^2 + 10y - 24 = 0 \quad \Longleftrightarrow \quad (y+12)(y-2) = 0.$$

We reject $y = -12$ because $y \geq 0$, so $y = 2$ and $x = 2^2 - 2 = 2$.

Check: $2 + 10\sqrt{4} - 22 = 0$. The only real root is $x = 2$.

Ask yourself What would have happened if we had kept $y = -12$?

It gives $x = 144 - 2 = 142$, and then $142 + 10\sqrt{144} - 22 = 240 \neq 0$. The value $y = -12$ is a root of the related equation $x - 10\sqrt{x+2} - 22 = 0$, which is what you'd be solving if you squared things up carelessly. The condition $y \geq 0$ is exactly what separates the two.

Part (ii)(b)

This time no substitution is suggested, so we look for one. The expression under the root is

$$2x^2 - 8x - 3 = 2(x^2 - 4x) - 3,$$

and $x^2 - 4x$ also appears outside the root. That's the link.

Ask yourself What should y be here: $x^2 - 4x$, or the whole square root?

Taking y to be the square root makes the equation a quadratic with no roots left in it. Let $y = \sqrt{2x^2 - 8x - 3}$, with $y \geq 0$. Then $y^2 = 2(x^2 - 4x) - 3$, so

$$x^2 - 4x = \frac{y^2 + 3}{2}.$$

Substituting this into the equation,

$$\begin{aligned} \frac{y^2+3}{2} + y - 9 = 0 \quad &\iff \quad y^2 + 2y - 15 = 0 \\ &\iff \quad (y+5)(y-3) = 0. \end{aligned}$$

Since $y \geq 0$, we need $y = 3$. Then

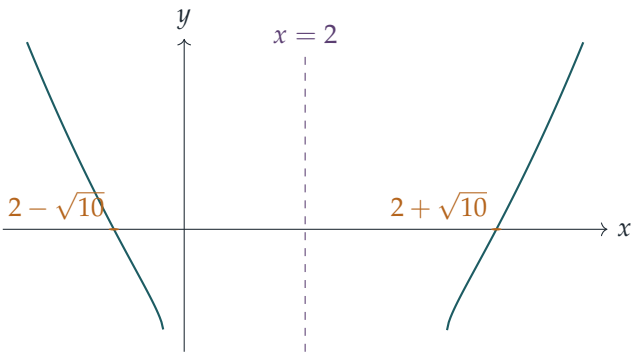
$$2x^2 - 8x - 3 = 9 \quad \iff \quad x^2 - 4x - 6 = 0 \quad \iff \quad x = 2 \pm \sqrt{10}.$$

Both values make $2x^2 - 8x - 3$ equal to 9, which is positive, so the square root is defined and equals 3. Hence both are genuine roots: the real roots are

$$x = 2 + \sqrt{10} \quad \text{and} \quad x = 2 - \sqrt{10}.$$

The pair is symmetric about $x = 2$, as it should be, because the equation depends on x only through $x^2 - 4x = (x-2)^2 - 4$.

The graph of the left-hand side shows this. It is defined only where $2x^2 - 8x - 3 \geq 0$, that is, for $x \leq 2 - \sqrt{5.5}$ or $x \geq 2 + \sqrt{5.5}$, and it crosses zero once on each branch.



Checks

- Part (i): $x = \frac{1}{2}(10 - 3\sqrt{11}) \approx 0.02506$, so $\sqrt{x} \approx 0.1583$ and $x + 3\sqrt{x} \approx 0.02506 + 0.47494 = 0.5$.
- Part (ii)(b): $x = 2 + \sqrt{10} \approx 5.1623$ gives $x^2 - 4x = 6$ and $2x^2 - 8x - 3 = 9$, so the left-hand side is $6 + 3 - 9 = 0$. The rejected value $y = -5$ would need $x^2 - 4x = 14$, and then the left-hand side is $14 + 5 - 9 = 10 \neq 0$.
- A numerical scan of each left-hand side over its whole domain finds exactly the roots above and no others.

Common slips

- Keeping the negative root of the quadratic in y. A square root is never negative, so that root does not give a solution.
- Squaring both sides straight away in (ii)(b). This gives a quartic, which is much harder, and it brings in extra roots that you would then have to test.
- In (ii)(a), forgetting that $x = y^2 - 2$, not y^2.

Beyond STEP

Introducing a new variable so that a hard equation becomes a familiar one is one of the most useful habits in mathematics. In (ii)(b) the equation depends only on $x^2 - 4x$, so it is a function of $(x - 2)^2$. Spotting a symmetry like this, and working with the quantity it leaves unchanged, comes up again in Galois theory, where you solve polynomial equations by finding expressions in the roots that are fixed by symmetries. Allenby's *Rings, Fields and Groups* is a gentle route into these ideas. For more problems of this kind, Siklos's *Advanced Problems in Mathematics* is excellent (and free).

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